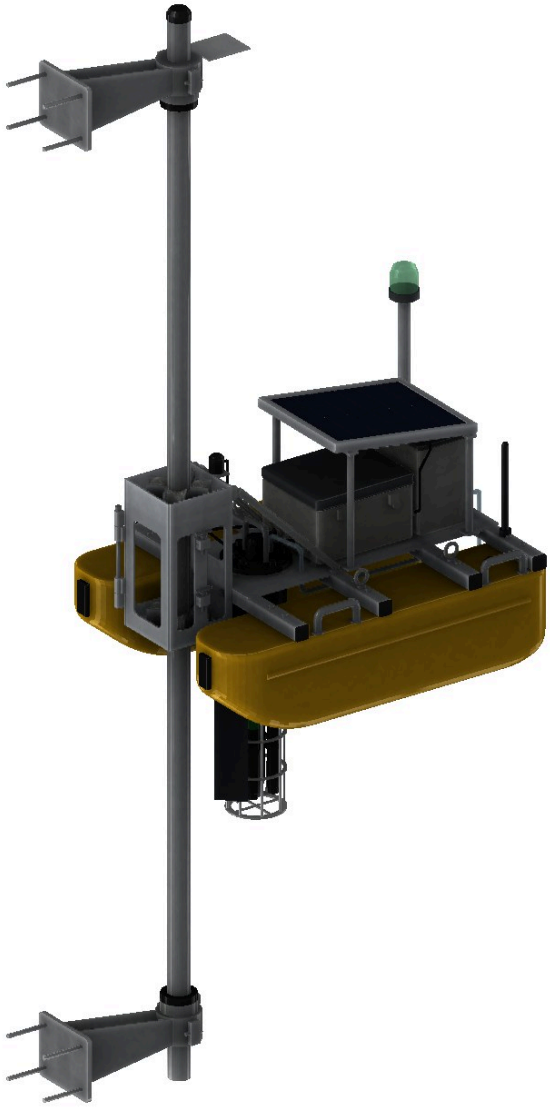
AquaSentinel R3

Full-size float: assembly manual



The full-size AquaSentinel: a twin-hull float that rides up and down a 2-inch guide pole fixed to a bridge pier, with its sensors in a well between the hulls. This manual builds it in 12 steps, from the pier brackets to the harness. Every picture is a view of our SOLIDWORKS 2026 assembly of the R3.

66 part numbers · 144 parts | 12 steps | 2 on site, 10 in the workshop | ≈ 16 h for two people (estimate)

Team Sanjivani Hero's Journey (AG26-1500$41), Sanjivani University, Kopargaon · Revision 1.0, 7 November 2026

Read this first

What this manual shows. The R3 exactly as it is in our SOLIDWORKS model, the design we submitted at the FDC. The R3 has not been built yet: this is the build plan, and the numbers are design estimates, not measurements.

> **PRODUCTION PLAN, NOT YET IN THE CAD**
> For the product we have chosen the Raspberry Pi electronics of our 1:5 live model, galvanised steel for the bridge parts instead of 316L, a bolted carriage and an HDPE deck (cost ≈ ₹48,000 per sentinel at 100 units). These changes will be modelled in SOLIDWORKS before the pilot. Each step points out where it is affected.

orange: parts added in this step grey: parts already fitted

Each step shows the new parts pulled out along their fitting direction (left) and fitted (right). Steps 1 and 2 happen on the pier; steps 3 to 12 build the float on the bank or in the workshop, and the float goes onto the pole at the end.

Key numbers

Float mass (CAD estimate)	**48.5 kg equipped · 43.6 kg dry carry**
Draft / freeboard	**117 mm / 143 mm**
Reserve buoyancy	**2.27 ×**
Guide pole	**2 in Sch 40S 316L, 2,978 mm long**
Bracket span	**2,678 mm**
Design travel	**2 m (site range to confirm)**
Rollers	**8 × UHMW-PE, two tiers 350 mm apart**
Design flood case	**1.5 m/s current + 500 N debris snag**

Safety

- Steps 1, 2 and the installation are done above water: life jackets, working in pairs, a throw line and the pier owner's approval.
- The float weighs about 44–49 kg: always a two-person lift on the grab handles, or a lift on the four eye bolts.
- Follow the anchor and adhesive suppliers' instructions and curing times.

Tools and fasteners

Tools

Hammer drill and anchor dispenser (on site) · torque wrench · metric spanners and hex keys · calipers · spirit level and plumb line · square and tape measure · leak-test pump · scales · cable-gland spanners.

Fasteners (all A4 / 316 stainless)

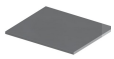
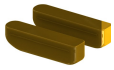
Part	Description	Qty	Grade / note	Step
AS3-312	Roller axle M10 + nyloc nuts	8	316 shoulder bolt, nylon isolation washers	5
AS3-313	Gate hinge pin D10	1	316 bar + R-clip	5
AS3-314	T-handle latch pin (tool-free)	2	316 bar + welded T-bar, lanyard eye	5
AS3-802	Lifting eye bolt M12 (DIN 580)	4		12
AS3-901	Bolt M8 x 60 A4 + washer	16	A4-70	4
AS3-902	Bolt M10 x 70 A4 + washer	2	A4-70	5
AS3-903	Screw M6 x 25 A4	6	A4-70	6
AS3-904	Chemical anchor M12 x 130 A4	8	A4 threaded rod + injection mortar	1
AS3-905	Bolt M10 x 40 A4 + washer	4	A4-70	2

TORQUE
Only the pole clamp bolts have a design torque: M10 at 35 N·m. For the other bolts use the A4-70 value from your fastener supplier. A full torque table is still to be added to the drawings.

STAINLESS ON ALUMINIUM
Use nylon washers, sleeves or insulating bushes wherever a stainless fastener or part touches the 6061 frame.

Parts list

From BOM_R3.csv · 66 part numbers · page 1 of 4

	Part	Description	Qty	Material	Step
	**AS3-101**	Pier bracket (weldment) - upper and lower identical	2	Stainless steel 316L	1
	AS3-102	Pole clamp cap	2	Stainless steel 316L	2
	**AS3-103**	Stage target plate	1	Stainless steel 316L	2
	AS3-111	Guide pole 2in Sch 40S SS316L	1	Stainless steel 316L	2
	**AS3-112**	Pole top cap	1	Acetal (POM) black	2
	AS3-113	Stop collar (split, clamp-on)	2	Stainless steel 316L	2
	AS3-114	Stop buffer ring (EPDM)	2	EPDM	2
	**AS3-201**	Pontoon hull (rotomoulded, foam filled)	2	Rotomoulded PE, RAL 1003 signal yellow	3
	**AS3-202**	Bow bumper pad	2	EPDM	3
	**AS3-203**	Hull vent / test plug	2	Acetal (POM) black	3
	**AS3-211**	Crossbeam SHS 40x40x3	4	Aluminium 6061-T6, anodised	4
	**AS3-212**	Beam end cap	8	Acetal (POM) black	4
	**AS3-213**	Equipment deck plate	1	Aluminium 6061-T6, anodised	4
	**AS3-301**	Guide carriage - fixed half (weldment)	1	Aluminium 6061-T6, anodised	5
	**AS3-302**	Guide carriage - opening gate (weldment)	1	Aluminium 6061-T6, anodised	5
	**AS3-311**	Concave guide roller UHMW-PE	8	UHMW-PE	5
	**AS3-312**	Roller axle M10 + nyloc nuts	8	Stainless steel 316 (brushed)	5
	AS3-313	Gate hinge pin D10	1	Stainless steel 316 (brushed)	5

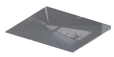
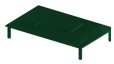
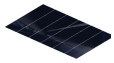
Parts list (continued)

From BOM_R3.csv · 66 part numbers · page 2 of 4

	Part	Description	Qty	Material	Step
	**AS3-314**	T-handle latch pin (tool-free)	2	Stainless steel 316 (brushed)	5
	**AS3-321**	Carriage brace struts (pair)	1	Stainless steel 316L	5
	**AS3-401**	Sensor well sleeve PE100 D160 SDR26	1	PE100 black	6
	**AS3-402**	Hatch flange ring (quarter-turn seat)	1	Stainless steel 316L	6
	**AS3-403**	Sensor guard cage	1	Stainless steel 316L	6
	**AS3-404**	V debris deflector	1	PE100 black	6
	**AS3-411**	Cartridge cap + wiper drive housing	1	Acetal (POM) black	11
	**AS3-412**	Cartridge lift handle	1	Stainless steel 316 (brushed)	11
	AS3-413	Cartridge spine + wiper shaft	1	Stainless steel 316L	11
	**AS3-414**	Probe holder spiders (x2)	1	Acetal (POM) black	11
	**AS3-415**	Optical wiper arm + brush pads	1	Stainless steel 316L	11
	**AS3-416**	Probe cables (in-well run)	1	Marine cable	11
	AS3-417	Sampler intake strainer + riser	1	Stainless steel 316 (brushed)	11
	**AS3-420**	Probe body (envelope)	4	Acetal (POM) black	11
	**AS3-421**	ID band - pH probe	1	pH ID band (blue)	11
	**AS3-422**	ID band - EC probe	1	EC ID band (orange)	11
	**AS3-423**	ID band - Optical DO	1	DO ID band (green)	11
	**AS3-424**	ID band - Turbidity probe	1	Turbidity ID band (magenta)	11

Parts list (continued)

From BOM_R3.csv · 66 part numbers · page 3 of 4

	Part	Description	Qty	Material	Step
	**AS3-501**	Pod mounting rails (pair)	1	Aluminium 6061-T6, anodised	7
	**AS3-511**	Power & control pod IP67 (body)	1	Polycarbonate RAL 7035	7
	**AS3-512**	Pod lid (hinged)	1	Polycarbonate dark grey	7
	**AS3-513**	Pod mounting plate	1	Aluminium 6061-T6, anodised	7
	**AS3-514**	LiFePO4 12.8 V 6 Ah with BMS	1	LiFePO4 pack	7
	**AS3-515**	Controller board (ESP32, RS485, RTC, microSD, LTE Cat-1)	1	PCB	7
	**AS3-516**	MPPT charge controller	1	Polycarbonate dark grey	7
	**AS3-521**	PV module 20 W (frame + backsheet)	1	Aluminium 6061-T6, anodised	8
	**AS3-522**	PV module 20 W (cells + glass)	1	Mono-Si PV cells + glass	8
	**AS3-523**	Canopy stanchions + hinge / pin blocks	1	Aluminium 6061-T6, anodised	8
	**AS3-601**	Autosampler cassette (insulated body)	1	PE white	9
	**AS3-602**	Autosampler lid (hinged, tamper-seal hasp)	1	Polycarbonate dark grey	9
	**AS3-603**	Bottle carousel (lift-out cassette)	1	Acetal (POM) black	9
	**AS3-604**	Sample bottle 250 mL HDPE (lab-supplied)	4	PE white	9
	**AS3-605**	Bottle cap (septum)	4	pH ID band (blue)	9
	**AS3-606**	Peristaltic pump 12 V (reversible)	1	Polycarbonate dark grey	9
	AS3-701	Beacon mast	1	Aluminium 6061-T6, anodised	10
	**AS3-702**	Jal-Deep beacon base	1	Polycarbonate dark grey	10

Parts list (continued)

From BOM_R3.csv · 66 part numbers · page 4 of 4

	Part	Description	Qty	Material	Step
	**AS3-703**	Jal-Deep 360 deg LED lens (green normal / red alert)	1	PC lens (green = normal)	10
	AS3-711	LTE antenna (whip, N-type)	1	Acetal (POM) black	10
	**AS3-712**	Stage sensor (waterproof ultrasonic, looking up)	1	Acetal (POM) black	10
	**AS3-801**	Grab handle (two-person carry)	4	Stainless steel 316 (brushed)	12
	**AS3-802**	Lifting eye bolt M12 (DIN 580)	4	Stainless steel 316L	12
	**AS3-811**	Cable harness (probes, stage sensor, PV, beacon)	1	Marine cable	12
	**AS3-812**	Sampler hoses (intake + purge-to-waste)	1	Clear PVC tubing	12
	**AS3-901**	Bolt M8 x 60 A4 + washer	16	Stainless steel 316L	4
	**AS3-902**	Bolt M10 x 70 A4 + washer	2	Stainless steel 316L	5
	**AS3-903**	Screw M6 x 25 A4	6	Stainless steel 316L	6
	AS3-904	Chemical anchor M12 x 130 A4	8	Stainless steel 316L	1
	AS3-905	Bolt M10 x 40 A4 + washer	4	Stainless steel 316L	2

The 12 steps at a glance



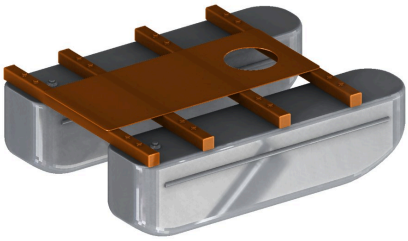
1 · Fix the pier brackets



2 · Fit the guide pole, stops and stage target



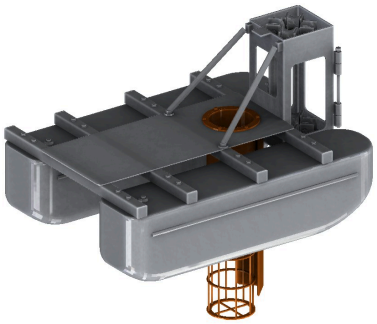
3 · Prepare the twin hulls



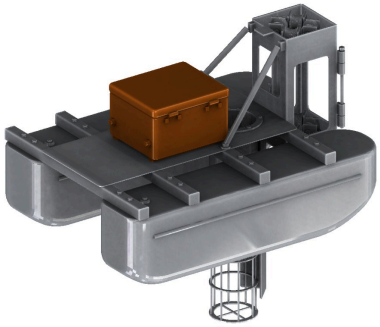
4 · Bolt on the crossbeams and the deck



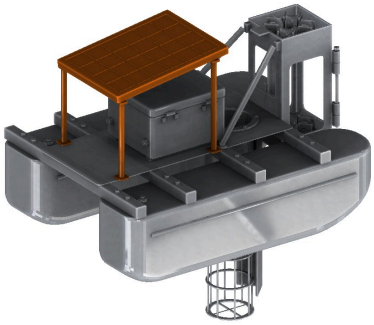
5 · Build and fit the guide carriage



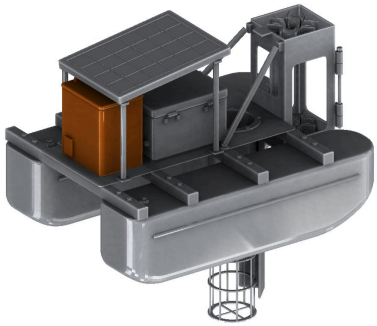
6 · Fit the sensor well, guard cage and deflector



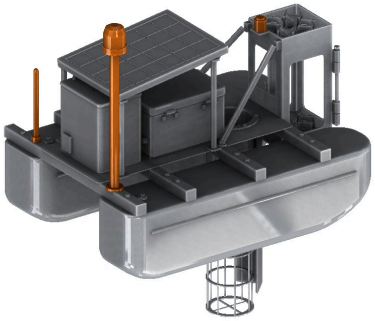
7 · Install the power and control pod



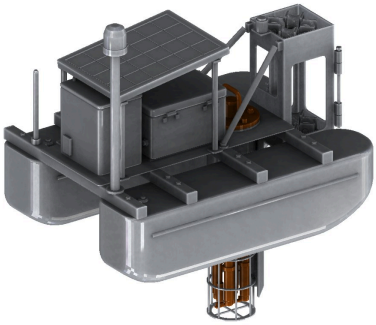
8 · Mount the solar canopy



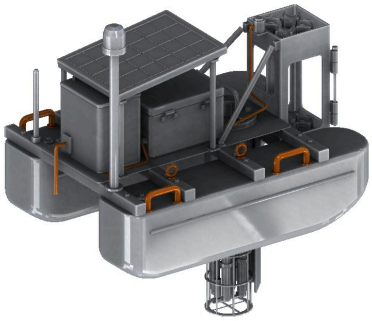
9 · Fit the autosampler



10 · Fit the beacon, antenna and stage sensor



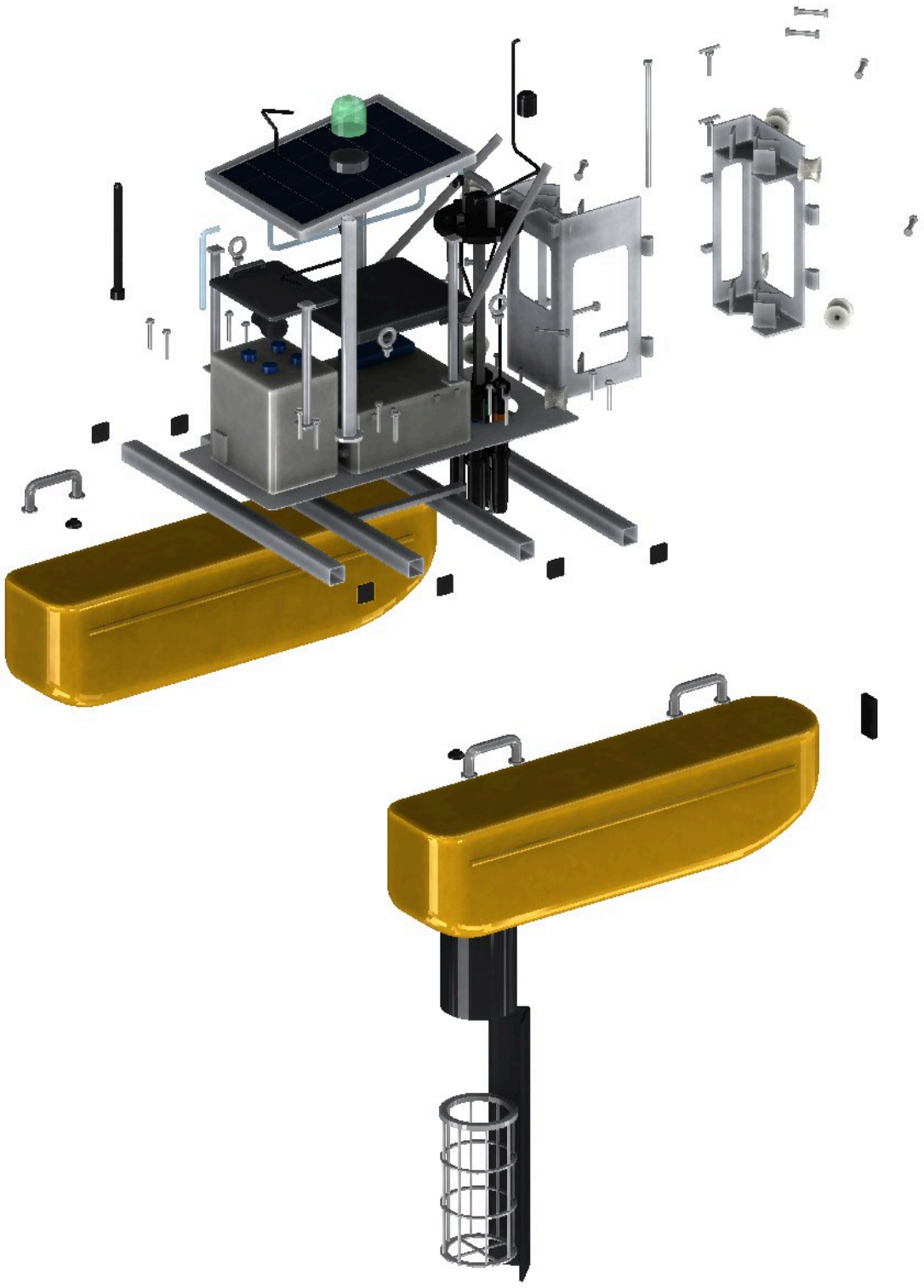
11 · Build and drop in the sensor cartridge



12 · Harness, hoses, handles and lifting eyes

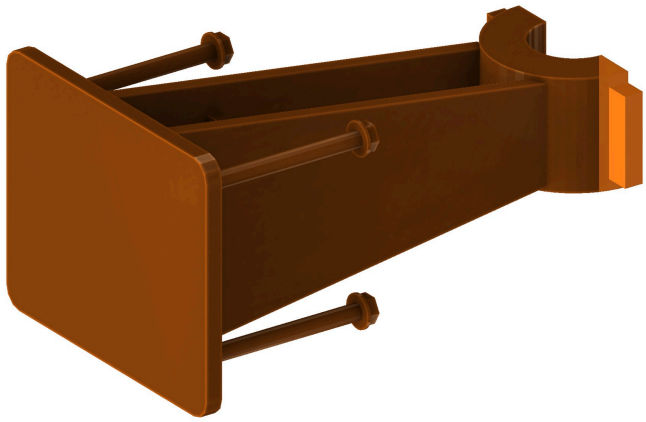
The float, exploded

Every part of the float pulled out along its fitting direction. Steps 3 to 12 put it back together, from the hulls up.



1 Fix the pier brackets

Where On site, at the bridge pier · **Joining** 4 chemical anchors per bracket, 160 × 120 mm pattern · **Tools** hammer drill, anchor dispenser, brush and blower, spirit level, plumb line



New parts pulled out along their fitting direction



Fitted



AS3-101 ×2
Pier bracket (weldment) - upper and lo

AS3-904 ×8
Chemical anchor M12 x 130 A4

1. Get written approval from the pier owner before drilling. Agree the anchor positions with them.
2. Mark the lower bracket first. Its back plate (220 × 180 mm) carries four M12 anchors on a 160 × 120 mm pattern.
3. Mark the upper bracket on the same vertical line, so that the two clamp seats share one axis. The design bracket span is 2,678 mm.
4. Drill, clean the holes (brush and blow), inject the mortar and set the A4 threaded rods, following the anchor supplier's instructions and curing time.
5. Fit both brackets, check them with a plumb line and tighten the anchor nuts once the mortar has cured.

CHECK

- Both clamp seats on one vertical line: the pole must run straight through both.
- Each arm holds the pole 360 mm off the pier face.

SAFETY

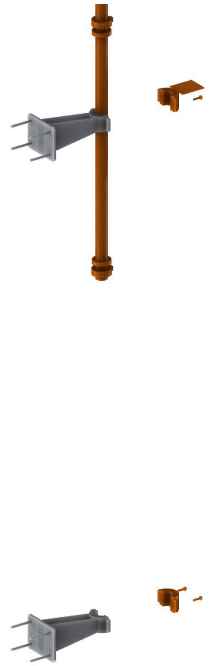
You are working above water. Wear a life jacket, work in pairs and keep a throw line ready.

OPEN DESIGN POINT

Anchor embedment, edge distances and the concrete grade have not been checked yet (interface M1). Design reaction: 430 N horizontal + 161 N pole weight.

2 Fit the guide pole, stops and stage target

Where On site · **Joining** split clamps, 2 × M10 per clamp at 35 N·m · **Tools** torque wrench, calipers, reamer if needed, hex keys



New parts pulled out along their fitting direction



Fitted



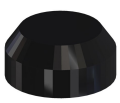
AS3-905 ×4
Bolt M10 x 40 A4 + washer



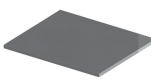
AS3-102 ×2
Pole clamp cap



AS3-111 ×1
Guide pole 2in Sch 40S SS316L



AS3-112 ×1
Pole top cap



AS3-103 ×1
Stage target plate



AS3-113 ×2
Stop collar (split, clamp-on)



AS3-114 ×2
Stop buffer ring (EPDM)

1. Measure the delivered pipe: outside diameter, ovality and straightness. The clamp bore is 60.7 mm reamed; the nominal pipe OD is 60.3 mm.
2. If the pipe is larger than the bore, ream the clamp caps to suit the measured pipe before going to site.
3. Lower the pole into both bracket seats. Fit the clamp caps and tighten the M10 bolts to 35 N·m. The stage target plate is welded to the upper clamp cap.
4. Slide on the EPDM buffer rings and clamp the stop collars with their M8 screws. They stop the float 20 mm short of the collars at both ends of its travel.
5. Press the black pole cap onto the top of the pole.

CHECK

- Pole vertical in both directions.
- Stop collars set so the float has its full design travel (2 m in this design).

DO NOT CUT EARLY

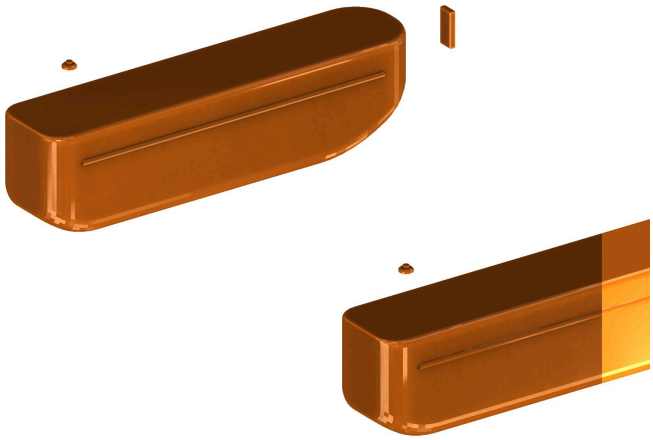
Do not cut the pole to length until the stage range of the site is confirmed. The 2 m range is an assumption.

OPEN DESIGN POINT

NPS 2 pipe can measure 59.5–61.1 mm with ovality (interface M2), and the clamp slip capacity is not calculated yet.

3 Prepare the twin hulls

Where Workshop · **Joining** bonded pads, threaded plugs · **Tools** leak-test pump, scales, flat assembly table, adhesive for EPDM



New parts pulled out along their fitting direction



Fitted



AS3-201 ×2
Pontoon hull (rotomoulded, foam filled



AS3-202 ×2
Bow bumper pad



AS3-203 ×2
Hull vent / test plug

1. Check each hull: 1,000 × 230 × 260 mm, wall 5 mm, eight moulded-in M8 inserts for the crossbeams.
2. Air-test each hull through the vent plug before the foam fill. The closed-cell foam keeps a holed hull afloat.
3. Weigh each hull and compare with the CAD estimate of 6.82 kg.
4. Bond an EPDM bumper pad on each bow and screw in the vent plugs.
5. Set both hulls on a flat table, inserts up, 540 mm centre to centre, bows level.

CHECK

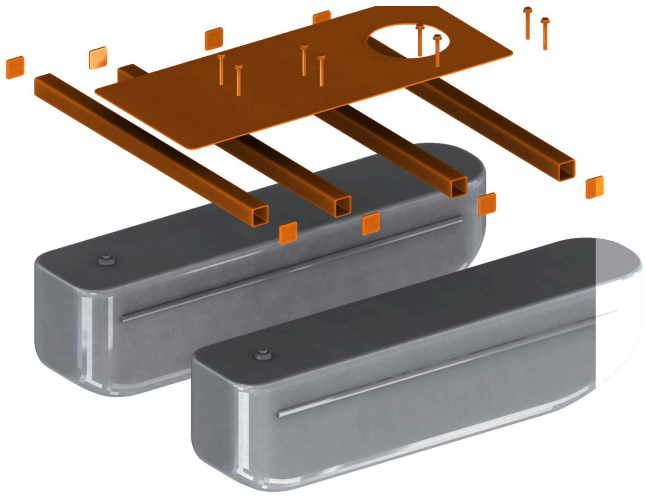
- No leak in the air test.
- Hulls parallel at 540 mm centres before the frame goes on.

OPEN DESIGN POINT

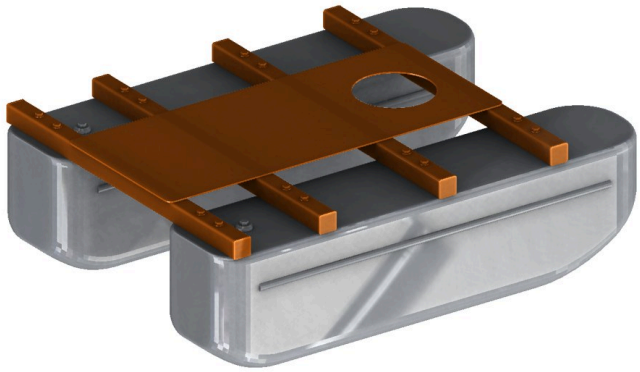
One mould makes both hulls, so one-sided handle inserts end up outboard on one hull and inboard on the other (interface M7). Mould inserts on both sides or mirror the layout.

4 Bolt on the crossbeams and the deck

Where Workshop · **Joining** 16 × M8 into the moulded-in inserts · **Tools** torque wrench, square, tape measure, nylon washers



New parts pulled out along their fitting direction



Fitted



AS3-211 ×4
Crossbeam SHS 40x40x3



AS3-212 ×8
Beam end cap



AS3-901 ×16
Bolt M8 x 60 A4 + washer



AS3-213 ×1
Equipment deck plate

1. Lay the four crossbeams across both hulls at x = 120, 380, 660 and 860 mm from the bow (pole) end.
2. Bolt each beam to the hull inserts with M8 × 60 A4 bolts. Put a nylon washer or sleeve under every stainless head on the aluminium.
3. Measure the two diagonals of the frame: they must be equal before you tighten fully.
4. Push the eight end caps into the beam ends.
5. Lay the deck plate on the beams, well opening over the hull centreline, and fix it.

CHECK

- Frame square (equal diagonals).
- Deck flat; well opening clear.

GALVANIC ISOLATION

Stainless fasteners on 6061 aluminium need nylon washers, sleeves or insulating bushes on every joint, or the aluminium corrodes.

5 Build and fit the guide carriage

Where Workshop · **Joining** M10 bolts, shoulder-bolt axles, hinge pin · **Tools** spanners, torque wrench, short offcut of the guide pole



New parts pulled out along their fitting direction



Fitted



AS3-312 ×8
Roller axle M10 + nyloc nuts



AS3-301 ×1
Guide carriage - fixed half (weldment)



AS3-302 ×1
Guide carriage - opening gate (weldmen



AS3-311 ×8
Concave guide roller UHMW-PE

AS3-314 ×2
T-handle latch pin (tool-free)



AS3-902 ×2
Bolt M10 x 70 A4 + washer

AS3-313 ×1
Gate hinge pin D10



AS3-321 ×1
Carriage brace struts (pair)

1. Fit the eight concave rollers on their M10 shoulder-bolt axles with nylon washers and nyloc nuts: four on the fixed half, four on the gate, in two tiers 350 mm apart.
2. Hang the gate on the fixed half with the D10 hinge pin and its R-clip. The gate swings 100° open.
3. Bolt the fixed half to crossbeam 1 with two M10 × 70 bolts.
4. Fit the pair of brace struts from the carriage to crossbeam 2.
5. Close the gate around a short offcut of the pole and lock it with both T-handle pins. Turn the offcut by hand.

CHECK

- Every roller spins freely.
- About 1.5 mm running clearance between each roller throat and the pole (roller centres 52 mm from the pole axis).

BEFORE ANY RIVER TRIAL

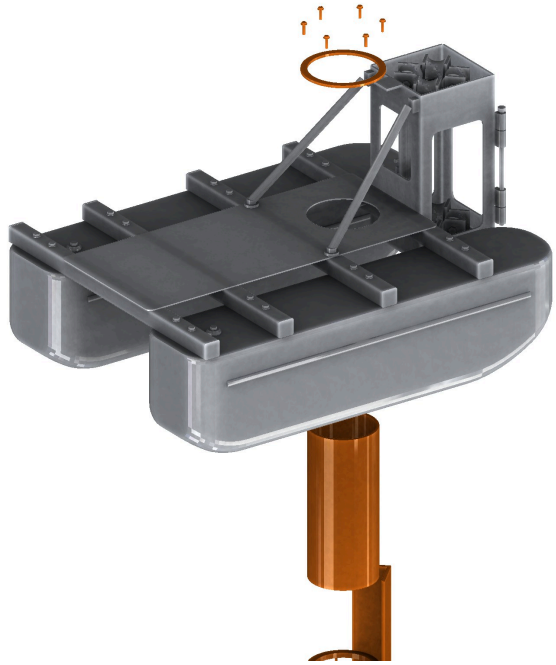
Current and debris push the float downstream, so the gate rollers carry up to about 1,107 N per tier. The hinge pin, latch pins, lugs and welds have not been checked yet (interface M5).

DESIGN MARGIN

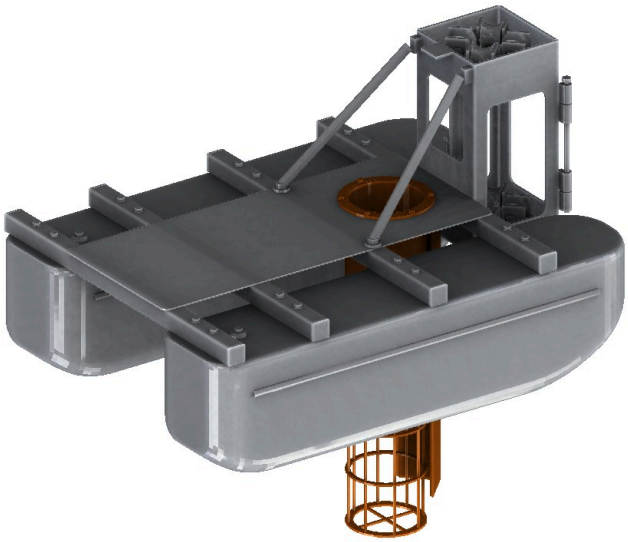
Anti-jam margin 8.62 at μ 0.05 and 1.72 with seized rollers (μ 0.25).

6 Fit the sensor well, guard cage and deflector

Where Workshop · **Joining** 6 × M6 into the deck, screws, bolts · **Tools** drill, hex keys



New parts pulled out along their fitting direction



Fitted



AS3-402 ×1
Hatch flange ring (quarter-turn seat)



AS3-903 ×6
Screw M6 x 25 A4



AS3-401 ×1
Sensor well sleeve PE100 D160 SDR26



AS3-403 ×1
Sensor guard cage



AS3-404 ×1
V debris deflector

1. Pass the PE100 sleeve down through the deck opening; it reaches 60 mm below the hull bottom datum.
2. Seat the 316L flange ring on the deck around the sleeve and fix it with six M6 × 25 screws.
3. Screw the guard cage to the bottom of the sleeve. Its rods protect the probes down to 380 mm below the datum.
4. Bolt the heat-bent V deflector under crossbeam 1, ahead of the well, point upstream.

CHECK

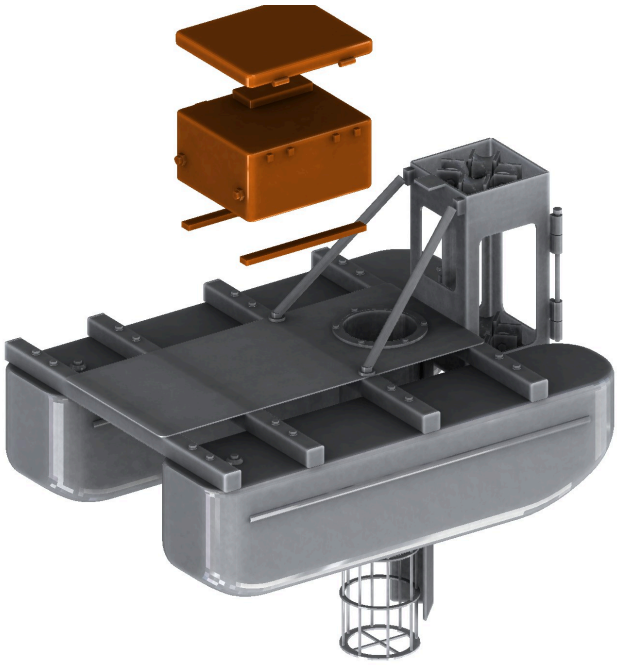
- Sleeve vertical and centred in the opening.
- Cage clear of the hulls and the carriage.

OPEN DESIGN POINT

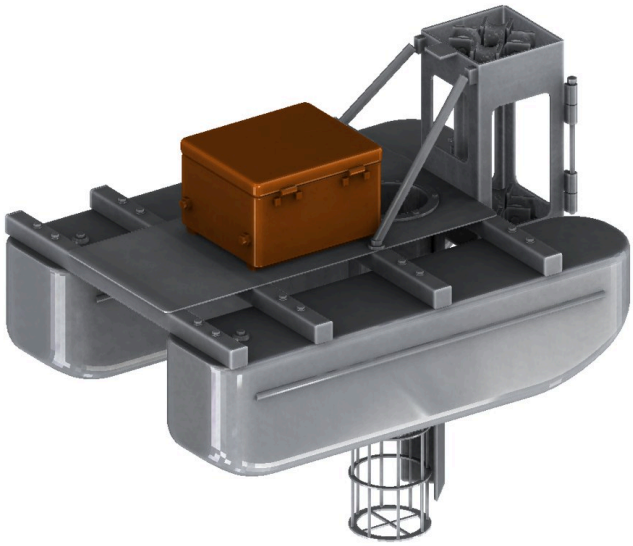
The sleeve-to-flange joint is not detailed yet (interface M8).

7 Install the power and control pod

Where Workshop · **Joining** rails and plate bolted to the deck · **Tools** screwdrivers, cable glands, multimeter



New parts pulled out along their fitting direction



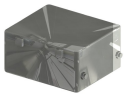
Fitted



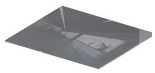
AS3-512 ×1
Pod lid (hinged)



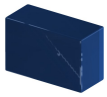
AS3-501 ×1
Pod mounting rails (pair)



AS3-511 ×1
Power & control pod IP67 (body)



AS3-513 ×1
Pod mounting plate



AS3-514 ×1
LiFePO4 12.8 V 6 Ah with BMS



AS3-516 ×1
MPPT charge controller



AS3-515 ×1
Controller board (ESP32, RS485, RTC, m

1. Bolt the two aluminium rails to the deck and fix the mounting plate on them.
2. Mount the IP67 enclosure on the plate. It is machined for ten M16 IP68 cable glands and a pressure-equalising vent.
3. Fit the battery, the charge controller and the controller board inside, with the battery low and central.
4. Close the hinged lid on its EPDM gasket.

CHECK

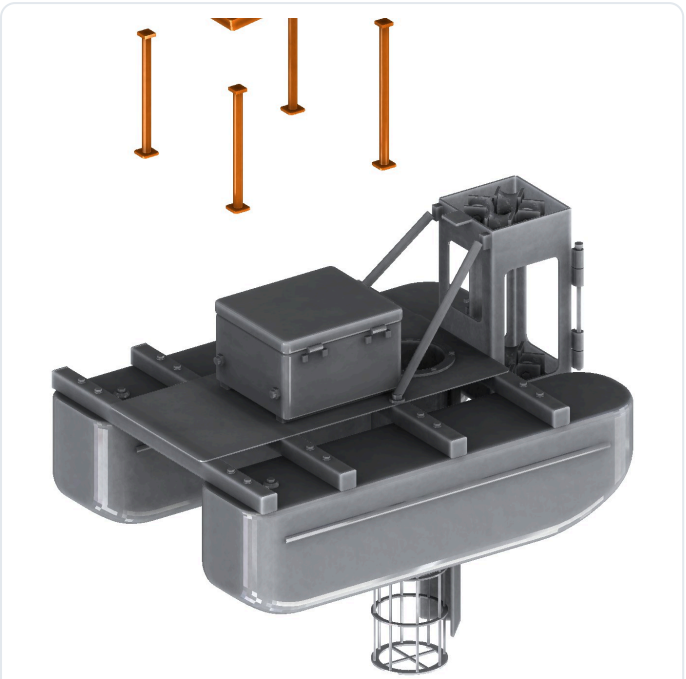
- Lid gasket seated all round.
- Immersion and 48 h soak test of the closed pod before it goes on the river (test plan).

PRODUCTION CHANGE, NOT YET IN CAD

The CAD pod holds an ESP32 controller. Our production plan uses the Raspberry Pi electronics of the 1:5 live model. The CAD enclosure (300 × 260 × 150) also differs from the costed catalogue box.

8 Mount the solar canopy

Where Workshop · **Joining** stanchions bolted to the deck; aft pair hinged, front pair quick-release pins · **Tools** spanners, spirit level



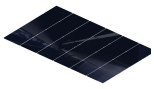
New parts pulled out along their fitting direction



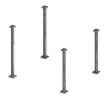
Fitted



AS3-521 ×1
PV module 20 W (frame + backsheet)



AS3-522 ×1
PV module 20 W (cells + glass)



AS3-523 ×1
Canopy stanchions + hinge / pin blocks

1. Bolt the four stanchions to the deck around the pod.
2. Fix the 20 W module on top at its 8° tilt. It also shades the pod.
3. Check the canopy hinges up on the aft pair once the front quick-release pins are pulled, so the pod lid can open.

CHECK

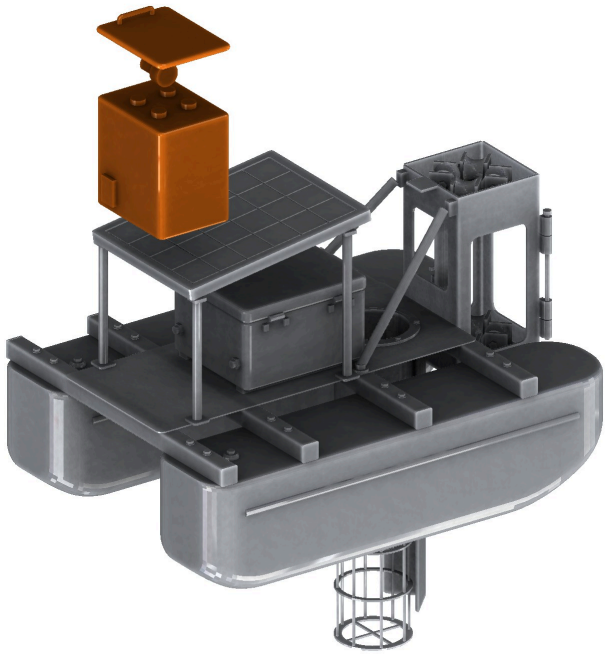
- Canopy clears the pod lid when hinged up.
- PV cable runs to a pod gland with a drip loop.

OPEN DESIGN POINT

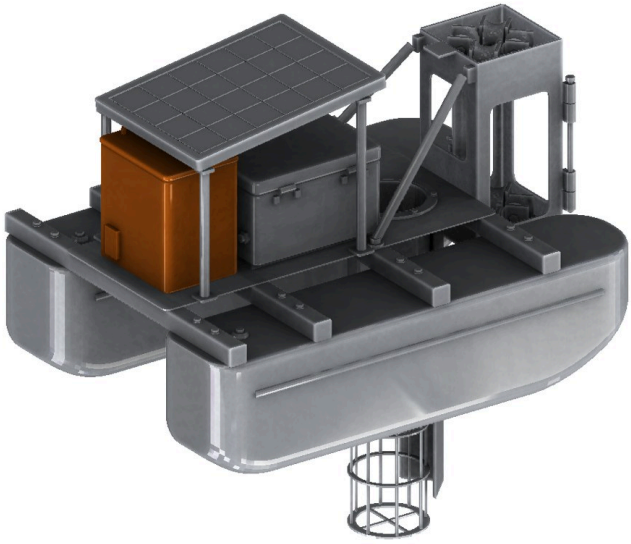
Wind load on the canopy has not been analysed (interface M12).

9 Fit the autosampler

Where Workshop · **Joining** cassette bolted to the deck · **Tools** screwdrivers



New parts pulled out along their fitting direction



Fitted



AS3-601 ×1
Autosampler cassette (insulated body)



AS3-602 ×1
Autosampler lid (hinged, tamper-seal h



AS3-604 ×4
Sample bottle 250 mL HDPE (lab-supplie



AS3-603 ×1
Bottle carousel (lift-out cassette)



AS3-605 ×4
Bottle cap (septum)



AS3-606 ×1
Peristaltic pump 12 V (reversible)

1. Fix the insulated cassette to the deck beside the pod.
2. Mount the peristaltic pump.
3. Load the four lab-supplied 250 mL bottles with septum caps into the carousel and drop the carousel into the cassette. The whole carousel lifts out, which keeps the chain of custody simple.
4. Close the lid and fit a tamper seal through the hasp.

CHECK

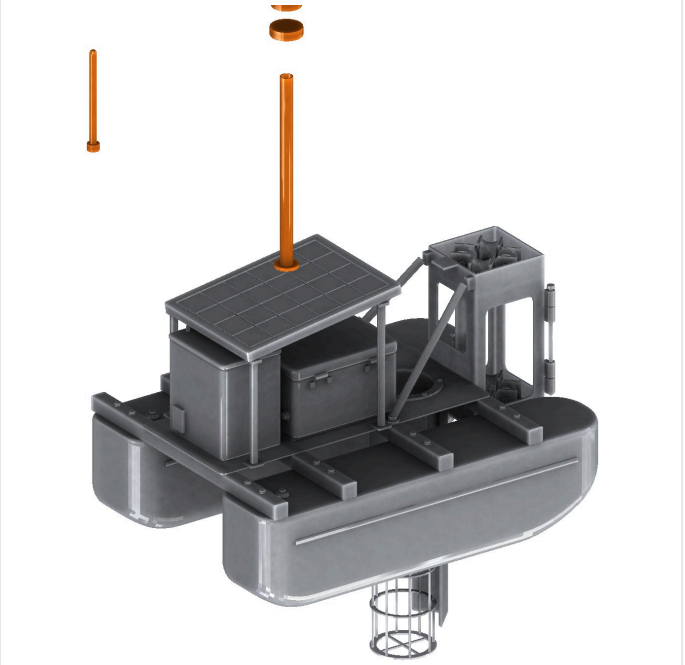
- Carousel lifts out and drops back in one piece.
- Lid seals and the hasp takes a seal.

CHANGED IN COSTING, NOT YET IN CAD

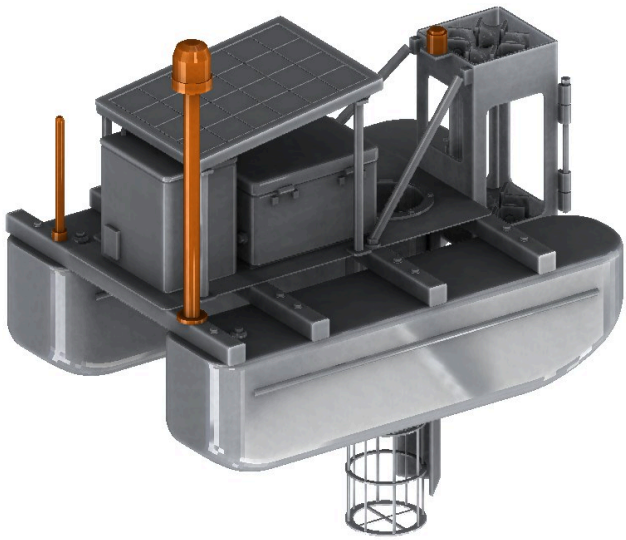
The costed version uses a Kamoer KPHM200 pump and two 12 V solenoid valves (sample and purge). Priming lift and line volume are still open (interface M13).

10 Fit the beacon, antenna and stage sensor

Where Workshop · **Joining** mast flange bolted to the deck · **Tools** spanners, cable glands

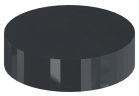


New parts pulled out along their fitting direction



Fitted

AS3-701 ×1
Beacon mast



AS3-702 ×1
Jal-Deep beacon base



AS3-712 ×1
Stage sensor (waterproof ultrasonic, I



AS3-703 ×1
Jal-Deep 360 deg LED lens (green norma



AS3-711 ×1
LTE antenna (whip, N-type)

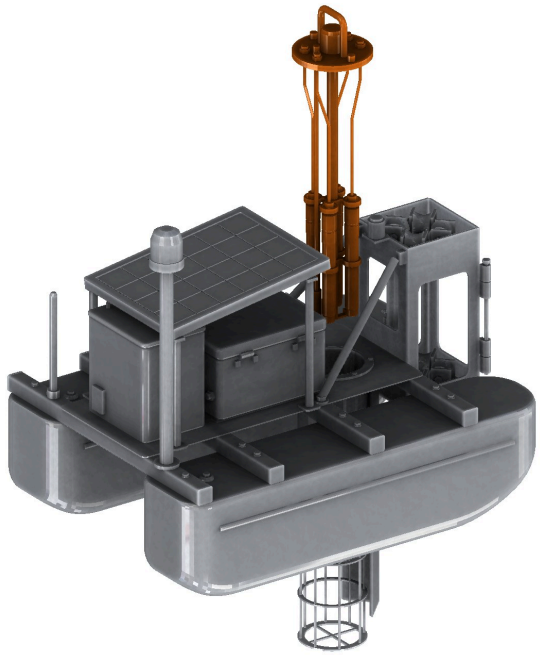
1. Bolt the beacon mast at the starboard aft corner of the deck and fit the beacon base and lens on top.
2. Fit the LTE whip antenna at the port aft corner.
3. Fit the waterproof ultrasonic sensor looking up at the stage target plate under the upper clamp. Its face sits 638 mm above the keel.

BEACON COLOURS
Red and green lights on water can be read as navigation marks at night. Agree the colours with the waterway authority.

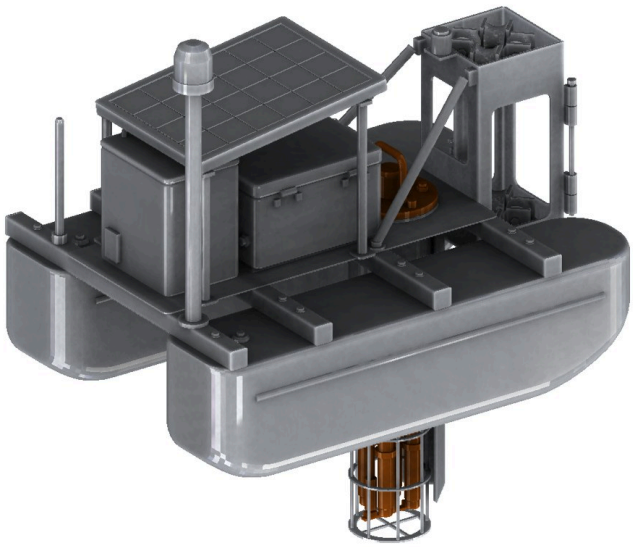
OPEN DESIGN POINT
At high water the range (146 mm) falls inside the sensor's 20–25 cm blind zone (interface M14). Remount it, or use a pressure transducer or a downward radar.

11 Build and drop in the sensor cartridge

Where Workshop, then on the float · **Joining** split collars in the spiders; cap seats on the flange with quarter-turn knobs · **Tools** hex keys



New parts pulled out along their fitting direction



Fitted



AS3-412 ×1
Cartridge lift handle

AS3-413 ×1
Cartridge spine + wiper shaft



AS3-414 ×1
Probe holder spiders (x2)



AS3-411 ×1
Cartridge cap + wiper drive housing



AS3-420 ×4
Probe body (envelope)

AS3-417 ×1
Sampler intake strainer + riser



AS3-421 ×1
ID band - pH probe



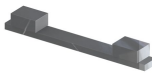
AS3-424 ×1
ID band - Turbidity probe



AS3-422 ×1
ID band - EC probe



AS3-423 ×1
ID band - Optical DO



AS3-415 ×1
Optical wiper arm + brush pads



AS3-416 ×1
Probe cables (in-well run)

1. Fix the two probe spiders on the spine (probe circle 90 mm).
2. Slide the four probes into the spiders with their tips level with each other, so the wiper passes over all of them.
3. Fit the colour bands: blue pH, orange EC, green DO, magenta turbidity.
4. Fit the wiper arm, the intake strainer and riser, and run the probe cables up through the cap glands.
5. Lift the cartridge by its handle, lower it 700 mm down the well and seat the cap on the flange with the quarter-turn knobs.

CHECK

- Probe tips level (wiper contact).
- Cartridge goes in and out without catching on the cage.

PLACEHOLDERS IN CAD

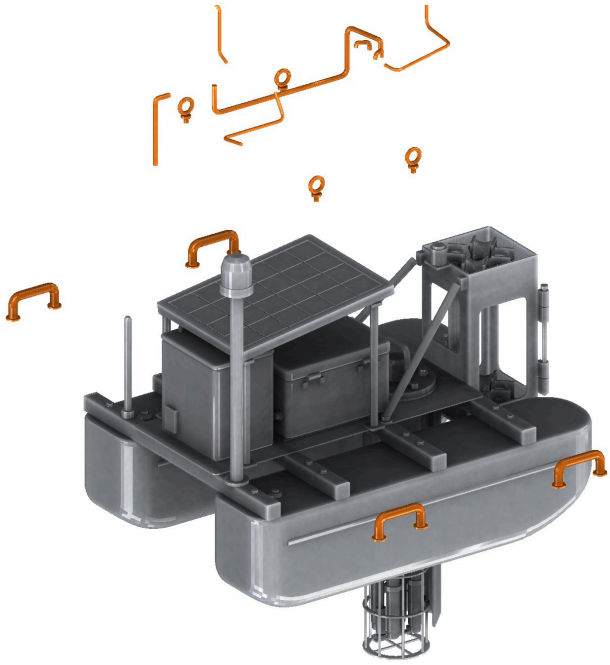
The probe bodies (D32 × 300) are envelopes, to be replaced with the supplier STEP files. The production set follows the live model's sensors (pH, TDS, turbidity, temperature).

OPEN DESIGN POINT

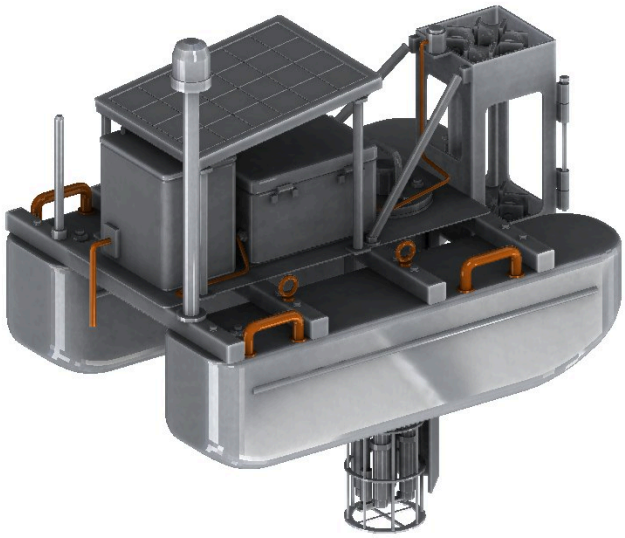
Cartridge locking and sealing are not detailed yet (interface M9). It weighs about 3.95 kg.

12 Harness, hoses, handles and lifting eyes

Where Workshop · **Joining** IP68 glands, hose connectors, threaded inserts · **Tools** cable ties, spanners



New parts pulled out along their fitting direction



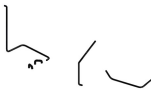
Fitted



AS3-802 ×4
Lifting eye bolt M12 (DIN 580)



AS3-801 ×4
Grab handle (two-person carry)



AS3-811 ×1
Cable harness (probes, stage sensor, P



AS3-812 ×1
Sampler hoses (intake + purge-to-waste

1. Run every cable into the pod through its IP68 gland, each with a drip loop.
2. Connect the 6 mm food-grade intake hose from the strainer to the pump, and the purge hose to the outlet aft of the deck.
3. Screw the four grab handles into the hull inserts.
4. Fit the four M12 lifting eyes for crane or davit lifts.

CHECK

- No cable can reach the rollers or the gate.
- Hoses free of kinks.

TWO-PERSON LIFT

The float weighs about 43.6 kg dry (cartridge out, bottles empty) and 48.5 kg fully equipped (CAD estimate). Always carry it with two people using the grab handles.

Views from our SOLIDWORKS 2026 assembly of the R3 (66 part files) · Team Sanjivani Hero's Journey, AG26-1500$41

Guide carriage (step 5)

Fixed half bolted to crossbeam 1 and braced to crossbeam 2; the gate hinges on a D10 pin and closes with two T-handle pins. Eight concave UHMW-PE rollers in two tiers 350 mm apart run on the 2-inch pole.



Sensor well and cartridge (steps 6 and 11)

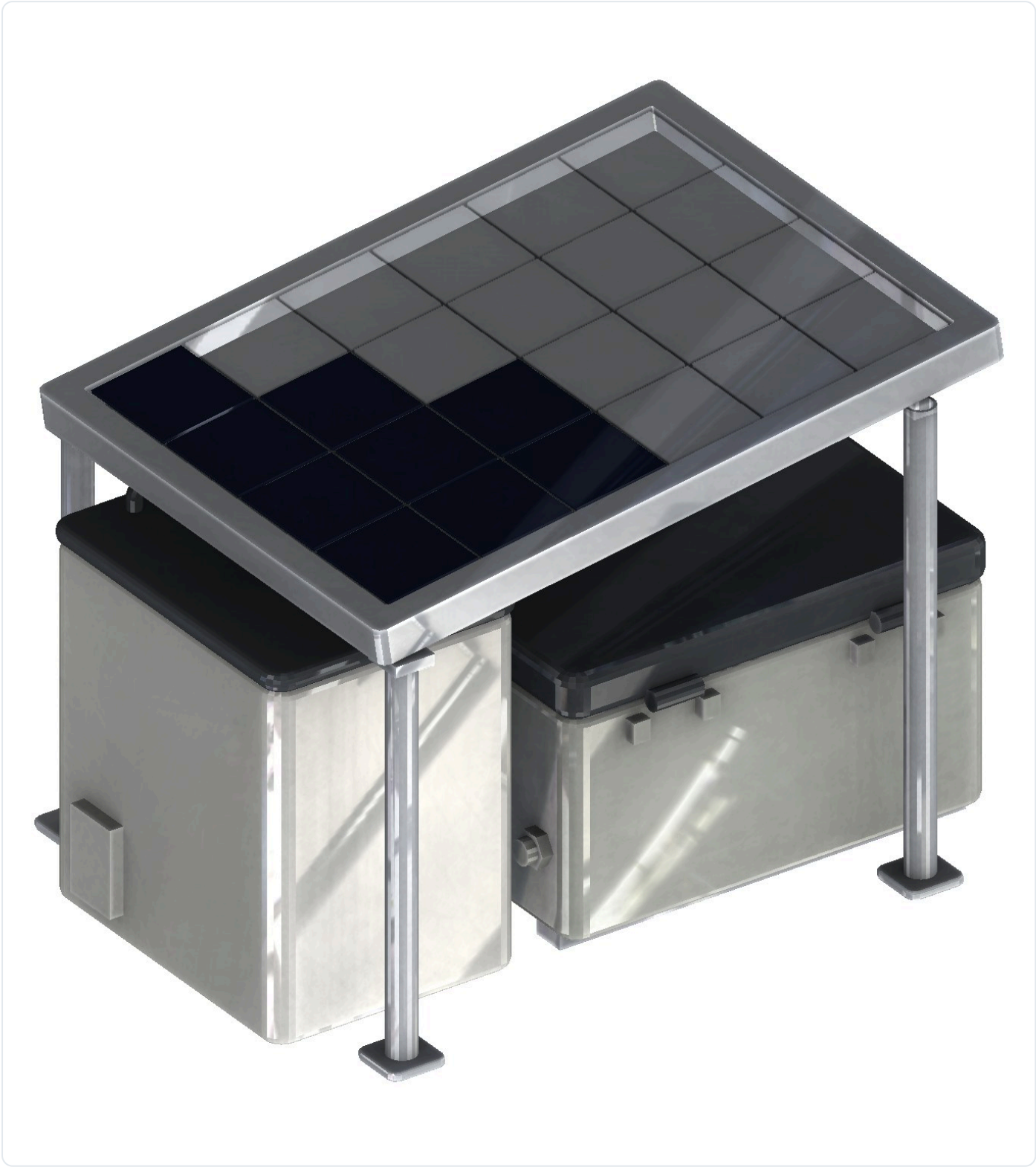
The PE100 sleeve passes through the deck; the 316L guard cage protects the probes below the hulls. The cartridge carries four probes, the wiper and the sample intake, and lifts out 700 mm by its handle for cleaning and calibration.



Views from our SOLIDWORKS 2026 assembly of the R3 (66 part files) · Team Sanjivani Hero's Journey, AG26-1500$41

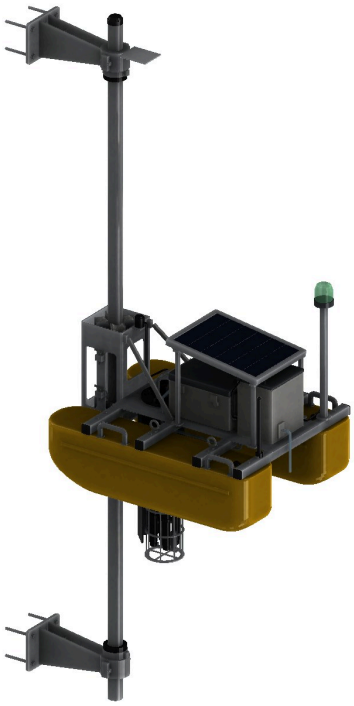
Pod, canopy and autosampler (steps 7, 8 and 9)

The IP67 pod holds the battery, the charge controller and the controller board; the 20 W canopy shades it and hinges up on the aft stanchions. The insulated autosampler keeps four 250 mL bottles on a lift-out carousel.

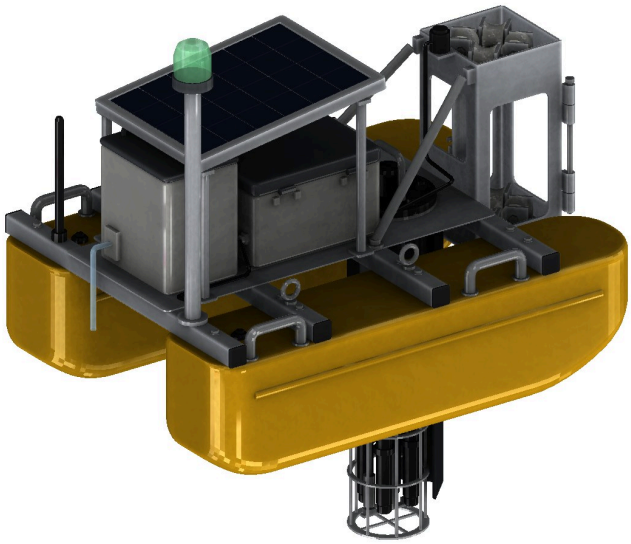


Views from our SOLIDWORKS 2026 assembly of the R3 (66 part files) · Team Sanjivani Hero's Journey, AG26-1500$41

Put the float on the pole



Float on the pole, bow side



The float, stern side

Installation

1. Carry the float to the pier with two people, on the grab handles.
2. Pull both T-handle latch pins and swing the carriage gate open.
3. Offer the carriage to the pole between the two stop collars and close the gate around it.
4. Push both latch pins fully home and check the float rides up and down freely on its eight rollers.
5. Commission the electronics (next page) before you leave the site.

Tests before the float works on the river

Stage	Tests (none done yet)
Incoming	Measure pipe OD, ovality and straightness; check material certificates (316L, 6061-T6); check probe addresses and baud rates.
Fabrication	Visual and dye-penetrant check of the bracket and carriage welds; passivation check; anodise thickness.
Hulls	Air leak test before foaming; mass and wall thickness; insert pull-out sample.
Float	Weigh the assembly (CAD estimate 48.5 kg). Tank test: draft 117 mm, freeboard 143 mm, trim, the full 2 m travel and roller friction on a pole rig.
Electronics	IP67 immersion of the pod; 48 h soak and leak test; bench energy test.
Commissioning	pH calibration with buffers; EC at 1,413 µS/cm; alarm and SMS end-to-end test; sampler purge and fill.

Open design points before the pilot

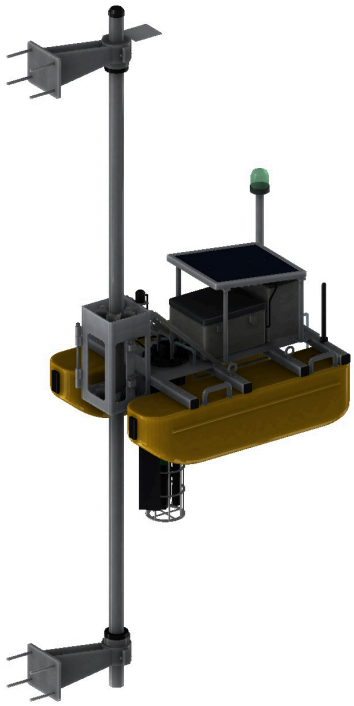
We are honest about what is not finished. These interfaces come from our 18 Oct design audit and must be closed before the R3 goes into a river.

ID	Interface	What is still open
M1	Pier ↔ bracket	Anchor embedment, edge distances, concrete grade, pier owner approval, impact and uplift loads.
M2	Bracket ↔ pole	Pipe OD tolerance (59.5–61.1 mm) against the 60.7 mm reamed bore; clamp slip capacity.
M3	Pole ↔ stops	Uplift on the upper stop above the top water level; friction capacity of one M8 screw.
M5	Gate ↔ fixed half	The gate carries the upstream rollers; hinge pin, latch pins, lugs and welds not analysed.
M6	Carriage ↔ float	Strength of the 6061 weldment heat-affected zone; galvanic isolation detail.
M7	Hull ↔ crossbeams	One mould but one-sided handle inserts; insert pull-out strength to confirm with the moulder.
M8	Deck ↔ sensor well	Sleeve-to-flange joint not detailed.
M9	Well ↔ cartridge	Locking, sealing and the lift of 3.95 kg over the water.
M10	Cartridge ↔ probes	Probe envelopes are placeholders; probe tips must be coplanar for the wiper.
M12	Canopy ↔ stanchions	Wind load on the canopy not analysed.
M13	Sampler ↔ water	Line volume and the priming lift from −335 mm to the pump at deck level.
M14	Stage sensor ↔ target	Range at high water inside the sensor's blind zone; a nearer reflector 48 mm to the side.

AND THE PRODUCTION CHANGES

Raspberry Pi electronics from the 1:5 live model, galvanised steel bridge parts, a bolted carriage, an HDPE deck, no wiper and a simpler bottle rack: about ₹48,000 per sentinel at 100 units. To be modelled in SOLIDWORKS before the pilot.

The finished R3



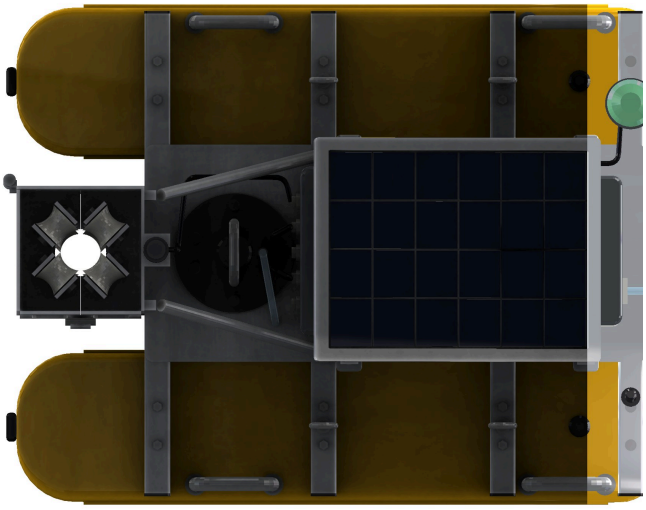
Front right



Front



Side



Float from above

Drawings: 05_R3_FULL_SIZE_CAD/04_Drawings · design checks: 05_R3_FULL_SIZE_CAD/05_Design_Checks · SOLIDWORKS assembly used for this manual: 05_R3_FULL_SIZE_CAD/03_SOLIDWORKS_2026_assembly_for_manual.